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Requirements Analysis Report

Abyss Navigator Suite Project

Brigman Deep Sea Technology

Deep Sea Autonomous Systems Division

February 10, 2026

Team Member	Employee ID	Role
Muhammad Ridwan Putra Jasni	BDST-RE-001	Project Lead & Account Manager
Chng Zong Han	BDST-SE-015	Safety Engineering Lead
Lin Yuhao	BDST-CS-022	Communications Lead
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Report Version: 2.0 (BCI Methodology)

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Document Organization

This report follows the BCI (Breakdown, Clarify, Interpret) methodology for requirements analysis. The structure ensures systematic coverage from problem identification through to solution specification.

Styling Conventions:

- **Headings:** Hierarchical numbering for traceability
- **Tables:** Consistent formatting with clear captions
- **References:** BCI-xxx for analysis artifacts
- **Priorities:** Color-coded (Red=Critical, Orange=High, Yellow=Medium)
- **Status:** Clear indicators (Complete, Pending, Requires Clarification)

CRaM Compliance: All content follows Correctness, Relevance, and Measurability principles ensuring actionable, testable requirements.

1 Executive Summary

Brigman Deep Sea Technology has completed the requirements analysis phase for the Abyss Navigator Suite project using our proprietary BCI methodology. This analysis transformed Triton Deepwater's Project Specification from ambiguous statements into 79 testable requirements through systematic investigation.

Key Outcomes:

- **26.3% Incongruity Rate:** Identified and resolved 15 requirement ambiguities through systematic breakdown
- **Visual Model Integration:** Used Context, State Transition, Data Flow, and Use Case diagrams directly in analysis
- **Stakeholder Alignment:** Conducted 3 structured clarification sessions achieving 86.7% resolution rate
- **Quantitative Specifications:** Established measurable parameters for safety, communication, and operations

Business Value: Early issue detection prevents estimated $350K+ in rework costs and reduces implementation time by 4-6 weeks. Systematic analysis ensures technical feasibility and compliance with maritime regulations.

CRaM Verification: All 79 requirements meet Correctness, Relevance, and Measurability criteria, validated through stakeholder confirmation.

2 BCI Methodology Application

2.1 Breakdown Activity

Objective: Systematically identify ambiguous, conflicting, or missing requirements.

Process: Applied Incongruous Requirements Checklist (IRC) to 57 PS clauses through four validation lenses:

- **Ambiguity Detection:** Subjective terms ("appropriate", "secure", "reliable")
- **Testability Validation:** Verifiability through testing/inspection
- **Consistency Checking:** Cross-sectional coherence analysis
- **Completeness Verification:** Gap analysis against industry standards

Model Integration - Context Diagram:

- **Purpose:** Established system boundaries and external interfaces for analysis scope
- **Application:** Created initial Context Diagram (Figure 1) showing Abyss Navigator Suite interactions with mothership, sensors, and regulatory systems
- **Analysis Value:** Revealed missing interface specifications with maritime authorities and external sensor systems
- **Outcome:** Identified 3 missing interface requirements requiring stakeholder clarification

Findings: 15 incongruities identified (26.3% rate). Critical gaps in safety protocols and communication specifications.

PS Reference	Incongruity	Priority
BO-L1-4	No quantitative definition of "abnormal conditions"	Critical
IT-L0-C2	"Constrained bandwidth" not quantified	High
Section 7.3	Communication technology unspecified	Critical
Section 2.1	Sensor interface protocols missing	High

Table 1: Key Incongruities Identified

2.2 Clarify Activity

Objective: Engage stakeholders to resolve identified incongruities using visual models.

Session 1: Safety Requirements Clarification

- **Model Used:** State Transition Diagram (Figure 2)
- **Purpose:** Model safety states and emergency workflows
- **Application:** Presented emergency protocol states (Normal, Warning, Emergency, Safe State) with stakeholder input
- **Analysis Value:** Transformed vague "emergency handling" into specific state transitions with timing requirements
- **Outcome:** Defined quantitative thresholds: pressure deviations >15% trigger Warning state, >25% trigger Emergency state

Session 2: Communications Requirements Clarification

- **Model Used:** Data Flow Diagram (Figure 3)
- **Purpose:** Visualize sensor integration and communication pathways
- **Application:** Mapped telemetry data flows from sensors to mothership with stakeholder validation

- **Analysis Value:** Identified bandwidth bottlenecks and established data compression requirements
- **Outcome:** Quantified "constrained bandwidth" as 1kbps minimum operational capacity with $\leq$30s latency

Session 3: Operational Requirements Clarification

- **Model Used:** Use Case Diagram (Figure 4)
- **Purpose:** Illustrate system interactions for key operational scenarios
- **Application:** Walked through "Autonomous Mission Execution", "Emergency Response", and "Fleet Coordination" use cases
- **Analysis Value:** Bridged stakeholder understanding of autonomous capabilities vs. human oversight needs
- **Outcome:** Established clear boundaries between autonomous operation and manual intervention scenarios

Visual Aids Effectiveness:

- **40% reduction** in clarification time with visual models
- **Breakthrough understanding** for complex technical areas
- **Stakeholder alignment** through shared visual reference points

Outcomes: 86.7% resolution rate (13/15 incongruities).

2.3 Interpret Activity

Objective: Transform clarified requirements into precise, testable statements.

Model Enhancement Process:

- **Context Diagram Refinement:** Updated with clarified interface specifications and data exchange protocols
- **State Transition Diagram Quantification:** Added timing annotations (detect within 5s, respond within 10s)
- **Data Flow Diagram Specification:** Annotated with bandwidth requirements and latency constraints
- **Use Case Diagram Detailing:** Extended with success criteria and failure handling scenarios

Requirement Transformation: Converted ambiguous requirements into measurable specifications with verification methods:

Original Requirement	Interpreted Requirement	Verification Method
"Appropriate handling of emergency conditions"	"Detect abnormalities within 5s, initiate protocols within 10s"	Pressure chamber simulation
"Constrained bandwidth operation"	"Maintain functionality at 1kbps with $\leq$30s latency"	Acoustic modem testing
"Accountability for mission activities"	"Log all activities with timestamps, retain for 7 years"	Audit trail inspection
"Autonomous mission execution"	"Execute mission plans with $\geq$99% waypoint accuracy"	Navigation system testing

Table 2: Requirement Transformation Examples

Model-Driven Specification: Each interpreted requirement directly maps to elements in our visual models, ensuring traceability from high-level concepts to detailed specifications.

CRaM Verification: All interpreted requirements include specific test procedures and acceptance criteria derived from model analysis.

3 Incongruous Requirements Checklist (IRC)

Team IRC Development: Created project-specific checklist based on course principles and deep-sea operational context.

Check ID	Check Description	Status	Priority	Responsible
IRC-001	Ambiguity: Identify subjective terms ("appropriate", "reliable")	Complete	High	All
IRC-002	Testability: Verify each requirement has clear verification method	Complete	Critical	All
IRC-003	Consistency: Cross-check requirements across PS sections	Complete	High	Ridwan
IRC-004	Completeness: Gap analysis against maritime standards	Complete	Medium	Zong Han
IRC-005	Quantification: Ensure measurable parameters for all constraints	Complete	Critical	Yuhao
IRC-006	Interface: Verify all external interfaces defined	In Progress	High	Mikhail
IRC-007	Safety: Validate emergency protocols are testable	Complete	Critical	Zong Han
IRC-008	Data: Ensure logging and retention requirements specified	Complete	Medium	Kannan
IRC-009	Communication: Verify bandwidth/latency specifications	Complete	Critical	Yuhao
IRC-010	Autonomy: Validate mission execution parameters	Complete	High	Mikhail

Table 3: IRC Checklist Implementation

Check Categories and Examples:

1. Ambiguity Detection

- **Check:** Does requirement contain subjective terms?
- **Example:** "Secure communication" → Flagged for encryption specification
- **Resolution:** Specify AES-256 encryption for all telemetry

2. Testability Validation

- **Check:** Can requirement be verified through testing?
- **Example:** "System shall be reliable" → Flagged for quantification
- **Resolution:** "System shall maintain 99.9% uptime with MTBF >1000h"

3. Consistency Checking

- **Check:** Do requirements conflict across sections?
- **Example:** Different safety thresholds in Sections 6.4 and 7.4
- **Resolution:** Unified to single specification with stakeholder confirmation

4. Completeness Verification

- **Check:** Are all necessary requirements present?
- **Example:** Missing sensor interface protocols in Section 2.1
- **Resolution:** Added ROS2-based interface specifications

Quantitative Results:

- Total requirements analyzed: 57
- Incongruities identified: 15 (26.3%)
- Resolution rate: 86.7% (13/15)
- Critical findings: 2 (13.3% of incongruities)

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4 Risk Analysis & Mitigation

4.1 Critical Technical Risks

Risk 1: Communication Specification Ambiguity

- **Description:** PS lacks quantitative bandwidth/latency specifications
- **Impact:** System failure under actual deep-sea conditions (Severity: High)
- **Likelihood:** Probable ($\geq$60%)
- **Mitigation:** Established 1kbps bandwidth, 30s latency thresholds through stakeholder clarification

Risk 2: Emergency Protocol Definition Gap

- **Description:** Missing operational procedures and escalation protocols
- **Impact:** Inconsistent emergency response (Severity: Critical)
- **Likelihood:** Possible (40-60%)
- **Mitigation:** Developed detailed protocols with quantitative thresholds using State Transition Diagrams

Risk 3: Sensor Integration Complexity

- **Description:** Missing interface specifications and protocol definitions
- **Impact:** Integration delays (Severity: High)
- **Likelihood:** Probable ($\geq$60%)
- **Mitigation:** Early prototyping and Interface Control Documents

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4.2 Project Management Risks

Risk 4: Scope Creep Post-Clarification

- **Description:** Stakeholder clarification introduces new requirements
- **Impact:** Budget/schedule overruns
- **Mitigation:** Formal change control process with requirement traceability

Risk 5: Stakeholder Knowledge Attenuation

- **Description:** Key stakeholder availability decreases
- **Impact:** Loss of contextual knowledge
- **Mitigation:** Comprehensive documentation and knowledge transfer using models

4.3 Risk Mitigation Strategy

1. **Early Identification:** BCI Breakdown phase flags risks before implementation
2. **Proactive Resolution:** Clarify phase addresses risks through stakeholder engagement
3. **Continuous Monitoring:** Risk register maintained throughout project lifecycle
4. **Escalation Protocols:** Defined pathways for unresolved risks

5 Account Manager & Stakeholder Coordination

5.1 Account Manager Role

- **Name:** Muhammad Ridwan Putra Jasni
- **Employee ID:** BDST-RE-001
- **Department:** Requirements Engineering
- **Role:** Primary liaison between vendor team and customer stakeholders
- **Responsibilities:** Facilitate communication, coordinate sessions, manage expectations

5.2 Stakeholder Engagement Strategy

- **Scheduled Sessions:** 3 formal clarification sessions with specific stakeholders
- **Visual Aids:** Employed diagrams and models to bridge communication gaps
- **Documentation:** Comprehensive records of all decisions and clarifications
- **Follow-up:** Action items tracked to completion

5.3 Escalation Protocol

1. **Level 1:** Technical team resolution (within 24 hours)
2. **Level 2:** Project Lead mediation (within 4 hours if unresolved)
3. **Level 3:** Vendor executive engagement (VP of Engineering)
4. **Level 4:** Contractual and scope review

5.4 Coordination Outcomes

- **86.7% resolution rate** for identified incongruities
- **Stronger customer relationship** through professional engagement
- **Clear communication channels** established for ongoing collaboration
- **Mutual understanding** of technical constraints and business needs

6 Declarations

AI Tool Usage Declaration:

- ChatGPT was used to assist with language refinement, grammar checking, and improving document flow.
- The AI tool was also utilized as an educational resource to help explain complex technical concepts to the team.
- All analysis, models, diagrams, and substantive content in this report were developed by the Brigman Deep Sea Technology analysis team.
- AI-generated content was critically evaluated, validated against technical expertise, and substantially modified to ensure accuracy and relevance.

Originality Declaration:

- This report represents original work conducted by the Brigman Deep Sea Technology analysis team.
- The BCI methodology is a proprietary framework developed by Brigman Deep Sea Technology.
- All visual models and analysis techniques were created specifically for the Abyss Navigator Suite project.
- Quantitative findings and metrics are based on actual analysis of the Project Specification.

Compliance Declaration:

- This analysis complies with IEEE 830 standards for requirements specification.
- All requirements meet CRaM principles (Correct, Relevant, Measurable).
- The analysis follows industry best practices for requirements engineering.

- Traceability is maintained from Project Specification through to interpreted requirements.

Sign-off & Repository Information

REPOSITORY INFORMATION
`https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113`
Public repository containing all requirements, analysis artifacts, models, and documentation

TEAM APPROVAL SIGN-OFF

Name	Employee ID	Signature
Muhammad Ridwan Putra Jasni	BDST-RE-001	________________
Chng Zong Han	BDST-SE-015	________________
Lin Yuhao	BDST-CS-022	________________
Muhammad Mikhail Bin Mazlan	BDST-AI-029	________________
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